



Emily J.

ID: LXBG

Gender: Female

Age: 79

Education: -

Primary Language: English

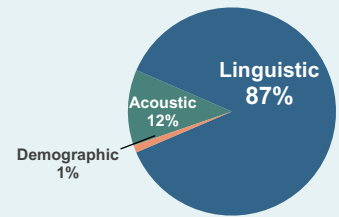
Probability of Predicted Outcome: 93.8%

Healthy

MCI

AD

Contribution of each modality to decision making



Listen to the voice!



0:27 / 0:30

Transcription

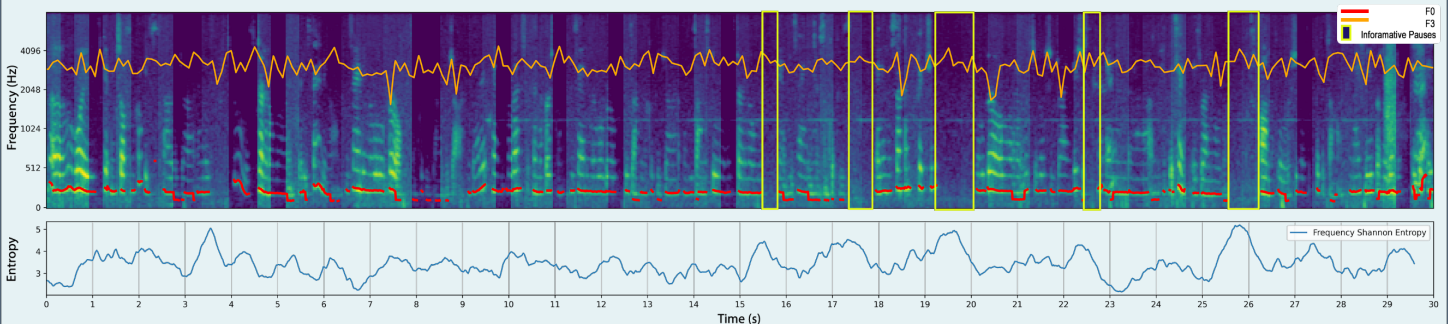
A little boy is getting himself some cookies out of the jar, and the stool is turning over on him that he's standing on. He's handing the little girl some cookies out of the jar. Is that it? Well, that's in the first one. He's handing the little girl cookies out of the cookie jar, and he's on the stool, and it's turning over, and he's handing her some cookies. And in the second picture, the little girl, she's drying dishes, and the sink is running over on the floor. She has pickets on her. Nice view from the window. Is that it?

Interpretation

The following language cues collectively suggest indicators of potential cognitive impairment:

- Frequent or repeated use of "Is," "that," "it," and "Well" suggests potential difficulties with word retrieval and memory.
- Disfluencies and filler phrases (e.g., "Well") indicate possible struggles in maintaining fluent and coherent thought.
- Reliance on vague terms (e.g., "that," "it") points to reduced semantic clarity, reflecting challenges in organized communication.
- Difficulties in sustaining a coherent narrative signal broader issues with cognitive processing and memory recall.

Spectrogram + SHAP Visualization



Interpretation

- Several instances of informative pauses (e.g., pauses before nouns) were detected.
- Signs of low energy levels were observed in the voice (see Spectrogram highlighted in green).
- Speech was less rhythmic, with some evidence of flat or monotonous segments (see Entropy curve, 8-15 seconds).
- The voice was unstable (standard deviation of Shimmer was high).
- Control over vocal folds was relatively acceptable, with some fluctuation observed (see F0 curve).
- Tongue-lip coordination for phoneme and syllable production was acceptable (see F3 curve).

Overall, there were several signs of cognitive impairment, suggesting the individual is likely cognitively impaired.

CONSIDERATIONS:

Please be advised that the sensitivity of this system is not 100%. A more comprehensive evaluation should include the individual's medical history and additional cognitive assessments.

To reduce the risk of cognitive status, evidence-based studies suggested:

- Having regular exercise
- Connecting with family/community
- Limiting alcohol intake

